Business Consulting Report

on

# Amazon Prime Air Drone Hub Location

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Dimension 1

# E-Commerce Business

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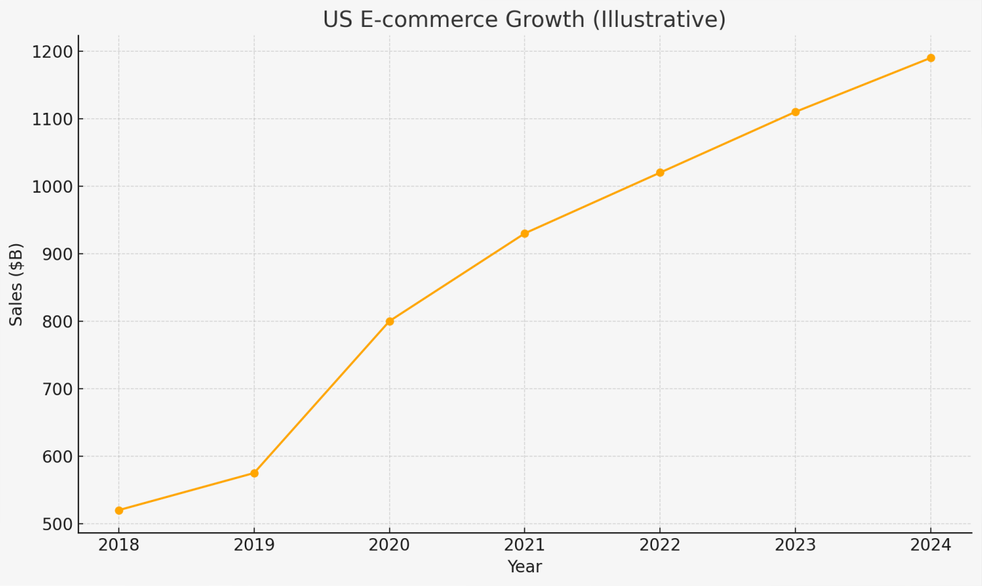
# Online Business & Demand Landscape

Demand is not uniform — cities must be viewed based on both location suitability and order intensity (high-frequency SKU movement)




**$8T**

Global online sales projected to reach **$8T by 2027**, rising from **$1.34T in 2014** — a 6× jump (Global trend)

**$1T**

U.S. online retail exceeded **$1T in 2024**, with e-commerce now 16.3% of total retail (U.S. trend)

**$110K**

MA demand reflects national patterns: **high income** (>$100K median) which indicates **high e-commerce frequency**, especially in Eastern MA (Boston region)

# Product Categories & Drone-Eligible Demand

Drone delivery viability begins with identifying the product categories and regions that drive the highest drone-eligible demand

**75%**

75% of Amazon orders **weigh ≤5 lbs**, making them ideal for MK30 drone delivery

**Drone-ready product categories**

- Fast-moving consumables (batteries, OTC, grooming)
- Light electronics (AirPods, cables, trackers)
- Small household essentials

**Demand tiers**

- **High demand regions**: Eastern MA (Boston, Newton, Waltham)
- **Moderate demand**: Worcester county
- **Lower but strategically important**: Western MA (Amherst, Northampton)

Dimension 2

# Drone Technology

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# Drone Technology: Capabilities & Cost Efficiency (MK30)

**Drone vs Truck - Cost, Speed & Environmental Impact**: Drones significantly outperform trucks on speed, cost per delivery and carbon impact

1

### Higher Avg Speed

Avg drone speed is ~50 miles/hour, enabling <15-minute 7–10 miles coverage loops

2

### Lower Cost per Delivery

Drone cost per delivery < $1 once scaled; truck delivery ~$3–$6 per stop (fuel, labor, congestion)

3

### Lesser CO2 Emissions

Drones reduce $CO_2$ emissions significantly: electric propulsion - cleaner last mile delivery

4

### MK30 Advantages

Quieter, handles rain/wind better, improved obstacle detection

5

### Long Term Financial Benefits

Initial investment higher, but long-term marginal cost drastically lower than trucks

6

### Improved Operations

Drone deliveries eliminate route inefficiencies by flying point-to-point, bypassing traffic delays and reducing overall delivery time variability

# Regulation, Feasibility & Constraints

1. **FAA Part 135 + MA zoning rules :** approvals required for flying over people, dense areas, and urban rooftops.

2. **Cold winters + coastal winds** reduce flight time; requires conservative radius planning

3. **Drones unsuitable for**: apartment blocks, dense tree areas, crowded city cores

4. **Environmental benefits**: reduced road traffic, low noise profile (MK30 quieter), minimized ground congestion

Understanding regulatory limits and environmental challenges is essential for planning feasible drone deployment zones

These constraints push early deployment toward suburban & rural areas

Dimension 3

# Population

# Population, Income & Demand Suitability

**Drone deployment should follow population gravity × income × SKU turnover, starting with safer, less dense Western/Central MA before scaling into high-complexity Eastern MA**

## Eastern Mass

**Primary demand corridor**

- Includes Boston, Middlesex, Norfolk, Essex, Plymouth counties.
- Concentrates roughly 60–65% of the state population, with some of the highest median household incomes (e.g., Middlesex & Norfolk > $120K)
- High income + high density - **strongest e-commerce and Prime demand**, ideal for later high-volume drone phases

## Central Mass

**Balanced density, mid-tier demand**

- Largely Worcester County, ~20% of MA population
- Median household income sits in the upper-middle range for the state.
- Provides a **natural midpoint** with solid yet manageable demand for early rollout.

## Western Mass

**Lower density, focused demand pockets**

- Hampden, Hampshire, Franklin, Berkshire counties together account for roughly **15–18%** of population.
- Incomes are generally lower than the east, but **student-driven micro-markets** (Amherst, Northampton, Five Colleges) create meaningful online order clusters.

# Landing Zone Feasibility & Airspace Limitations

Most drone-friendly
Predictable landing zones
Fewer obstructions
Strong adoption potential

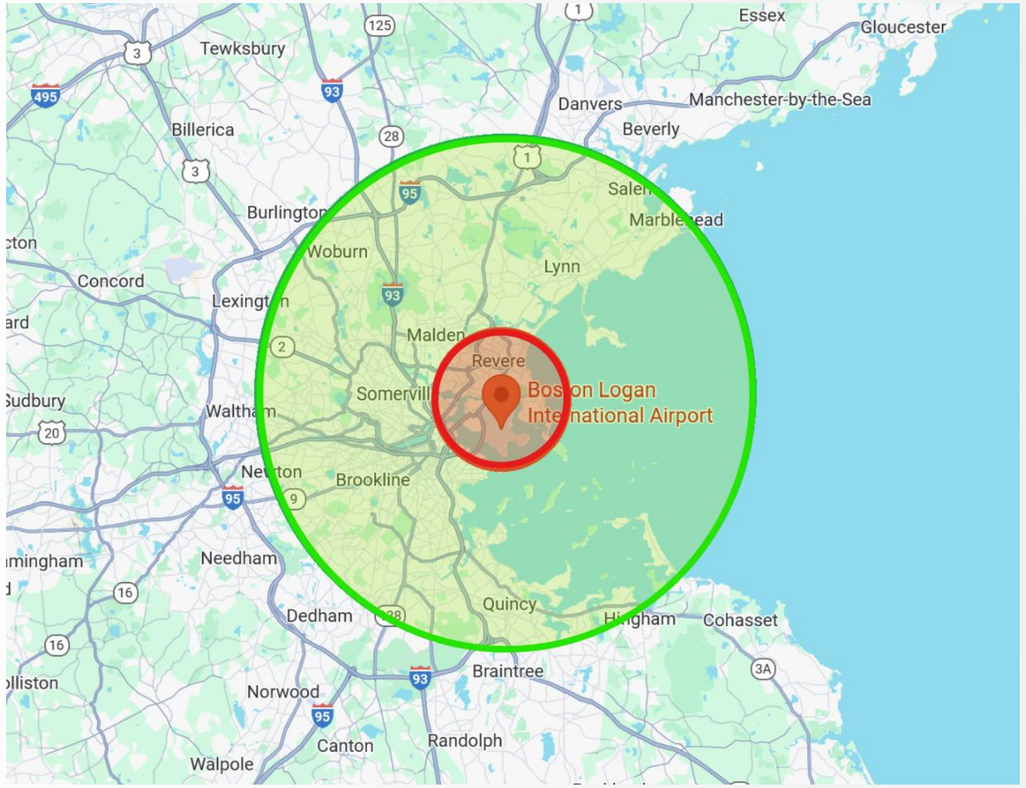
**1** **Suburban Areas**

Wide-open spaces and safer operations
Ideal for early-phase testing

**2** **Rural Areas**

Restricted by Logan Airport's Class B airspace
Congested and tall buildings

**3** **Urban Areas**




- **Red Zone** indicates prior FAA approval flying zones
- **Green Zone** indicates on-demand (only on orders) flying zone

**Suitability** = demand density × airspace feasibility × landing environment

Dimension 4

# Location Analysis

# Hub Network Design (COG + Strategic Choices)

## Western Hub

**Holyoke/South Hadley**

Initial COG in the **Springfield–Holyoke** region, **Holyoke/South Hadley** selected because:

- Western MA generates lower demand, allowing **safe early-phase rollout**.
- **Proximity to Five College region** (UMass, Amherst, Smith, Mount Holyoke, Hampshire) - high student-driven demand.
- **Existing Amazon Fulfilment Centre** in Holyoke improves operational integration.

## Central Hub

**Worcester**

- **Geographic midpoint** providing strong east–west connectivity
- Suburban landing feasibility
- Steady medium demand

## Eastern Hub

**Waltham (North/West Belt) & Quincy (South Shore)**

•**Avoid Logan Class B airspace** while covering the largest population clusters in MA.

•Support rapid fulfillment for high-density zones with **strong Prime penetration**

# Rollout Plan & Future Enhancements

**Phase 1**

**Western & Central MA**

Begin in somewhere between **South Hadley-Holyoke** and **Worcester** to operate in **safer, less congested airspace** and **build maturity before scaling**

**Phase 2**

**Eastern MA**

Expand to **Waltham** and **Quincy** to penetrate Boston's **high-demand corridor** once reliability is proven

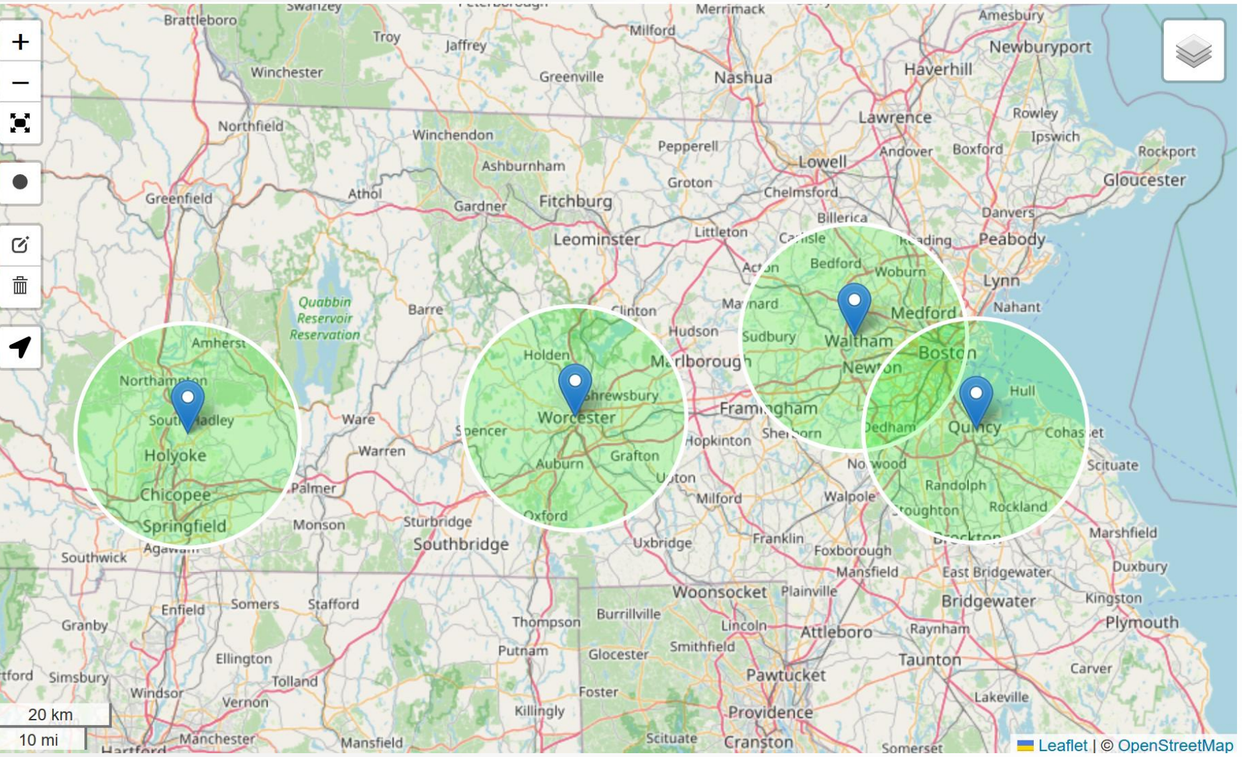
**Phase 3**

**Network Expansion & Optimization**

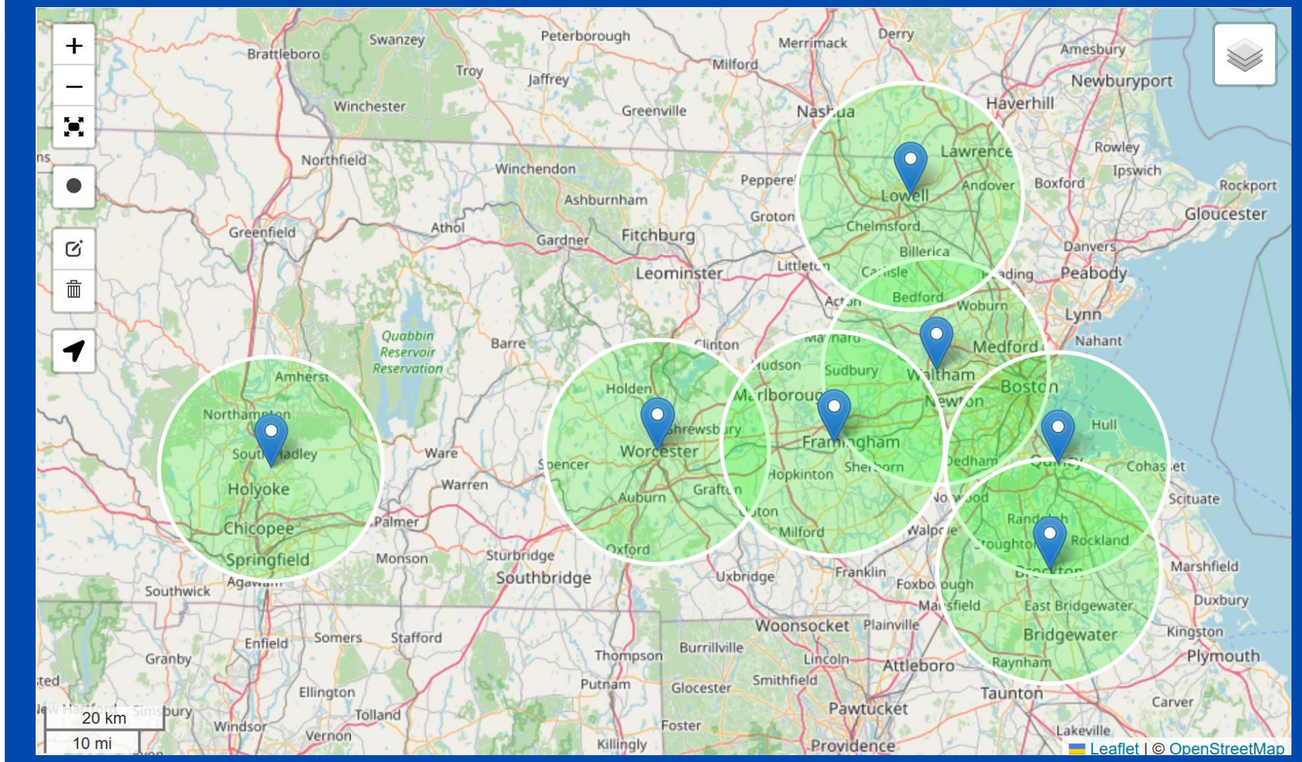
Launch micro-hubs in **Lowell, Framingham**, and **Brockton** and **improve routing, inventory**, and **support** as demand expands

1. **Automated drone towers** for high-volume throughput
2. **Nighttime low-noise delivery** leveraging MK30 acoustics
3. Partnerships with **MIT** & **Harvard** to establish FAA-approved campus delivery points

# After Phase 2




# After Phase 3



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